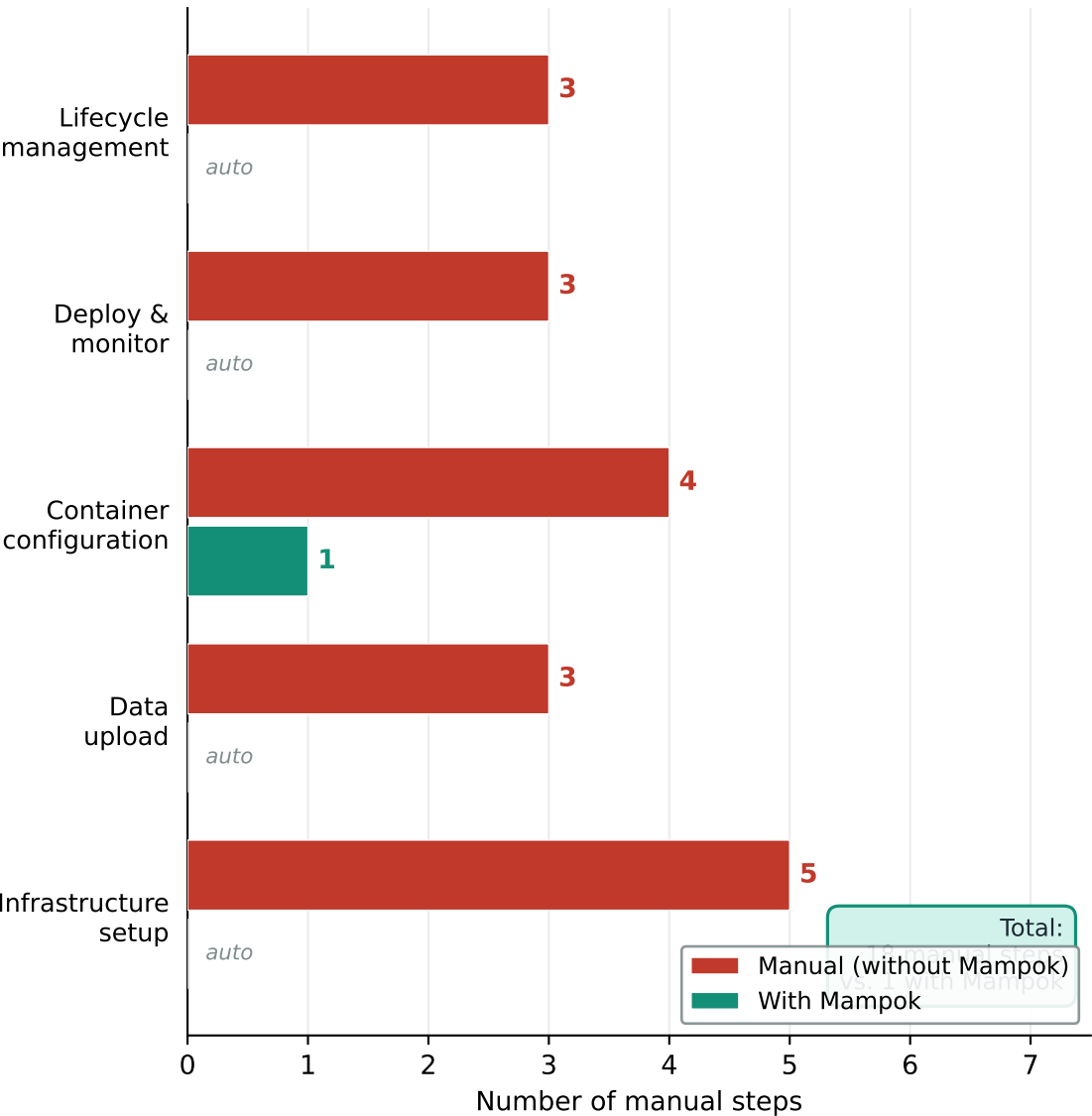


Steps required per task



Step-by-step comparison: deploying CellXGene

Without Mampok		With Mampok	
✗	kubectl create namespace	—	Automated by Mampok
✗	kubectl create secret (S3)	—	Automated by Mampok
✗	Create kubeconfig entry	—	Automated by Mampok
✗	Configure TLS / cert-manager	—	Automated by Mampok
✗	Set up Ingress controller	—	Automated by Mampok
✗	aws s3 mb s3://bucket	—	Automated by Mampok
✗	aws s3 cp data.h5ad s3://...	—	Automated by Mampok
✗	Verify upload integrity	—	Automated by Mampok
✗	Write Deployment YAML	✓	Write mamplan.json
✗	Write Service YAML	—	Automated by Mampok
✗	Write Ingress YAML	—	Automated by Mampok
✗	Configure resource limits	—	Automated by Mampok
✗	kubectl apply -f *.yaml	—	Automated by Mampok
✗	kubectl get pods (polling)	—	Automated by Mampok
✗	kubectl describe (debug)	—	Automated by Mampok
✗	Write cleanup cron job	✓	mampok stop-expired (cron)
✗	kubectl delete resources	—	Automated by Mampok
✗	aws s3 rm (data cleanup)	—	Automated by Mampok
Total: 18 manual steps		Total: 1 user step + 7 automated	